

Task 2: Branching Commands — Hands-On

List all branches in your repo

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git branch
* master
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |
```

Create a new branch called feature-1

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git branch feature-1
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git branch
  feature-1
* master
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |
```

Switch to feature-1

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git switch feature-1
Switched to branch 'feature-1'
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git branch
* feature-1
  master
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git status
On branch feature-1
nothing to commit, working tree clean
ubuntu@ip-172-31-30-155:~/devops-git-practice$
```

Create a new branch and switch to it in a single command — call it feature-2

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git switch -c feature-2
Switched to a new branch 'feature-2'
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git branch
  feature-1
* feature-2
  master
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git status
On branch feature-2
nothing to commit, working tree clean
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |
```

Try using git switch to move between branches — how is it different from git checkout?

I don't see any difference between checkout and switch both the command does the same thing but git checkout is used in creating branches and when applied on files removes your local changes without a way of undoing it.

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git checkout feature-1
Switched to branch 'feature-1'
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git switch main
fatal: invalid reference: main
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git switch master
Switched to branch 'master'
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git checkout feature-2
Switched to branch 'feature-2'
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |
```

Make a commit on feature-1 that does not exist on main & Switch back to main — verify that the commit from feature-1 is not there

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git branch
* feature-1
  feature-2
  master
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git status
On branch feature-1
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  add.sh

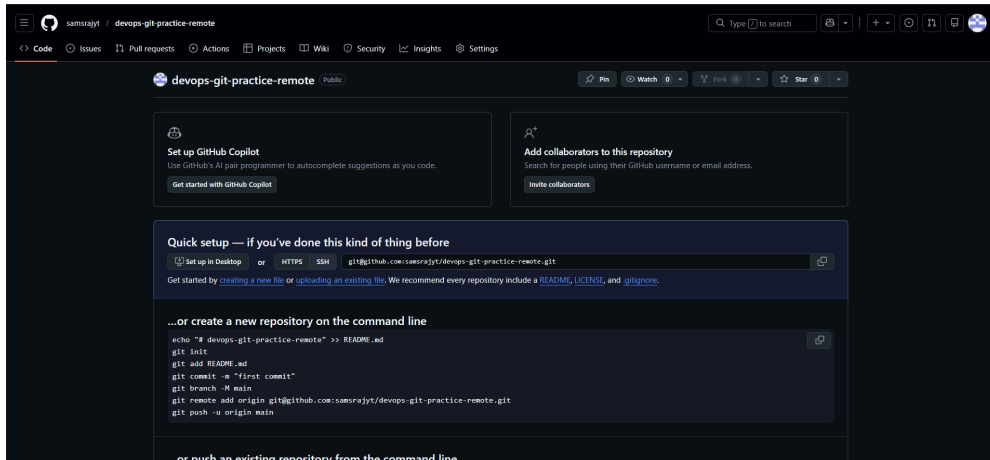
nothing added to commit but untracked files present (use "git add" to track)
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git add add.sh
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git commit -m "Added add.sh to feature-1 branch"
[feature-1 23c83be] Added add.sh to feature-1 branch
 1 file changed, 11 insertions(+)
 create mode 100755 add.sh
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git status
On branch feature-1
nothing to commit, working tree clean
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git log --oneline
23c83be (HEAD -> feature-1) Added add.sh to feature-1 branch
3ce969d (master, feature-2) Added index.html file
c3d31f2 Added script greet.sh
c4ba176 My First Shell Script
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git switch main
fatal: invalid reference: main
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git switch mater
fatal: invalid reference: mater
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git switch master
Switched to branch 'master'
ubuntu@ip-172-31-30-155:~/devops-git-practice$ ls -ltr
total 12
-rwxrwxr-x 1 ubuntu ubuntu 33 Mar 12 15:16 hello-world.sh
-rwxrwxr-x 1 ubuntu ubuntu 67 Mar 12 15:21 greet.sh
-rw-rw-r-- 1 ubuntu ubuntu 52 Mar 12 15:23 index.html
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |
```

Delete a branch you no longer need

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git branch -d feature-2
Deleted branch feature-2 (was 3ce969d).
ubuntu@ip-172-31-30-155:~/devops-git-practice$
```

Task 3: Push to GitHub

Create a new repository on GitHub (do NOT initialize it with a README)



Connect your local devops-git-practice repo to the GitHub remote

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git remote add origin git@github.com:samsrajyt/devops-git-practice-remote.git
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git remote -v
origin  git@github.com:samsrajyt/devops-git-practice-remote.git (fetch)
origin  git@github.com:samsrajyt/devops-git-practice-remote.git (push)
```

Push your main branch to GitHub

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'github.com:samsrajyt/devops-git-practice-remote.git'
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git branch
* feature-1
* master
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git push origin master
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 843 bytes | 843.00 KiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To github.com:samsrajyt/devops-git-practice-remote.git
 * [new branch]      master -> master
ubuntu@ip-172-31-30-155:~/devops-git-practice$
```

The screenshot shows a GitHub repository page for 'devops-git-practice-remote' by user 'samsrajyt'. The repository is public and has 1 branch (master) and 0 tags. The commit history shows 3 commits, with the latest commit 'Added index.html file' by 'samsrajyt' 18 hours ago. The repository contains files: 'greet.sh', 'hello-world.sh', and 'index.html'. A 'README' section is visible with a prompt to 'Add a README'. The right sidebar shows repository statistics: 0 stars, 0 watching, 0 forks, 0 releases, 0 packages, 0 contributors, and a language distribution of 65.8% Shell and 34.2% HTML.

devops-git-practice-remote Public

master 1 Branch 0 Tags

Go to file t Add file <> Code

samsrajyt Added index.html file 3ce969d · 18 hours ago 3 Commits

greet.sh	Added script greet.sh	18 hours ago
hello-world.sh	My First Shell Script	18 hours ago
index.html	Added index.html file	18 hours ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

This is a repository for practice purpose

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors

No contributors

Languages

Shell 65.8% HTML 34.2%

Suggested workflows

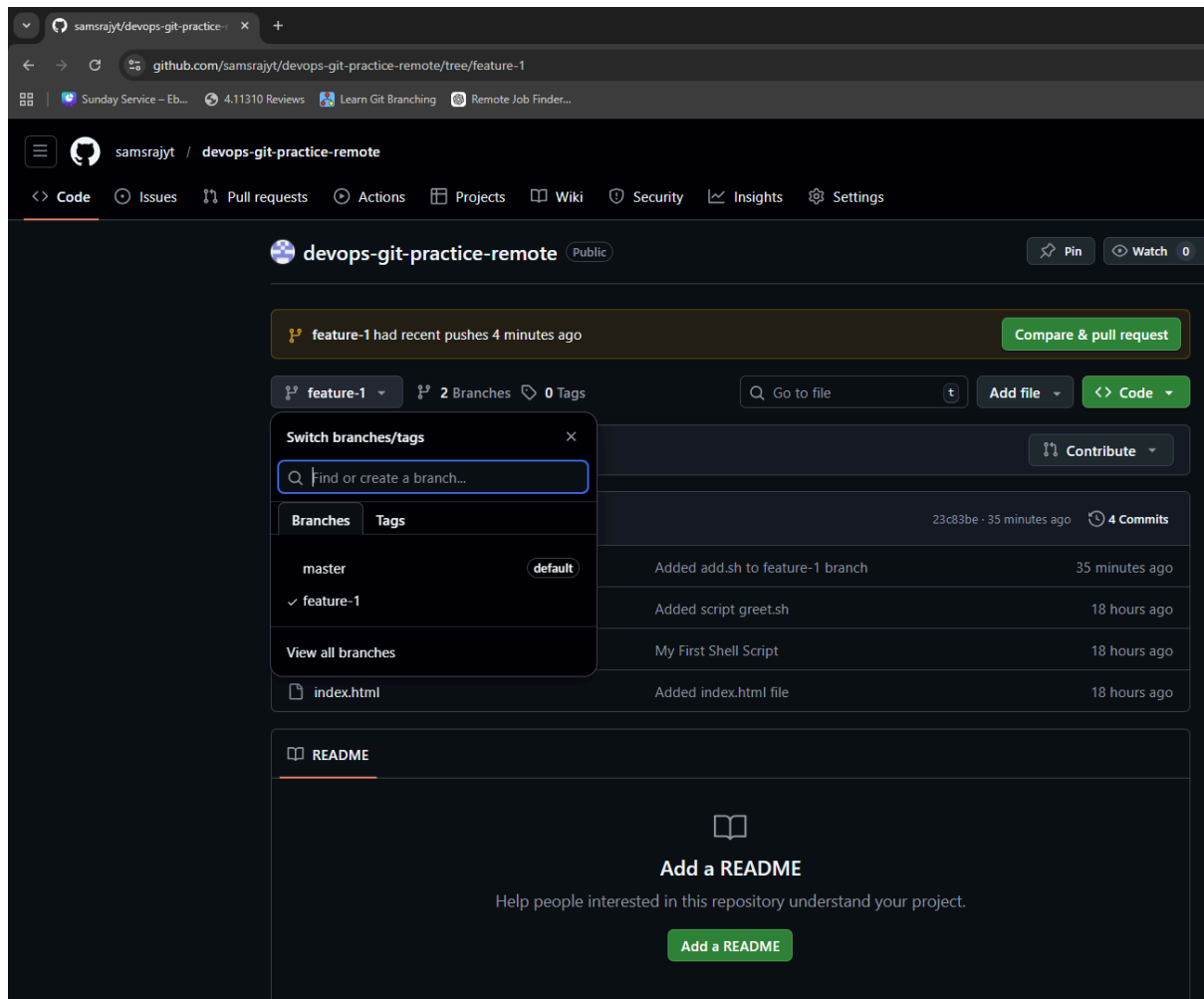
Push feature-1 branch to GitHub

The screenshot shows the GitHub web interface for a repository named 'devops-git-practice-remote' by user 'samsrajyt'. The repository is public and has 1 branch (master) and 0 tags. A notification bar at the top indicates that 'feature-1' had recent pushes 2 seconds ago, with a 'Compare & pull request' button. Below the notification, there is a table of recent commits. The first commit by 'samsrajyt' added an 'index.html' file 18 hours ago. Below the commit table, there is a 'README' section with a prompt to 'Add a README' to help people understand the project.

Commit	Author	Message	Time
3ce969d	samsrajyt	Added index.html file	18 hours ago
		Added script greet.sh	18 hours ago
		My First Shell Script	18 hours ago
		Added index.html file	18 hours ago

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git branch
feature-1
* master
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git switch feature-1
Switched to branch 'feature-1'
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git push origin feature-1
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 392 bytes | 392.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:   https://github.com/samsrajyt/devops-git-practice-remote/pull/new/feature-1
remote:
To github.com:samsrajyt/devops-git-practice-remote.git
* [new branch]      feature-1 -> feature-1
ubuntu@ip-172-31-30-155:~/devops-git-practice$
```

Verify both branches are visible on GitHub



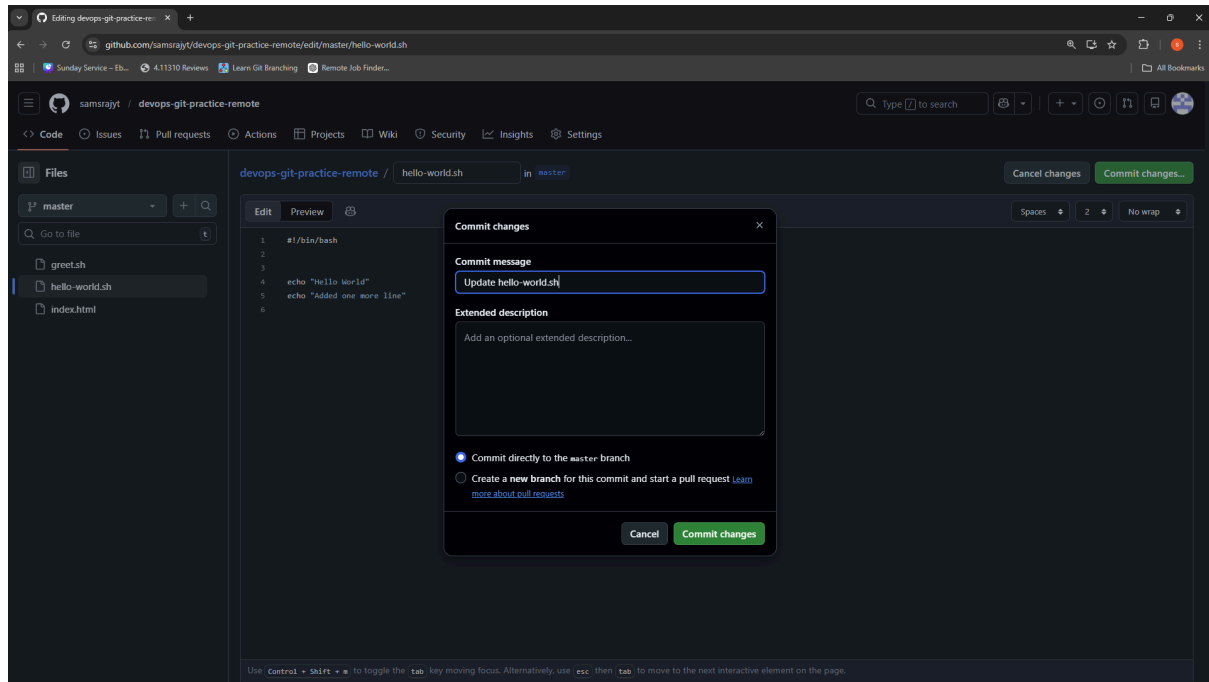
What is the difference between origin and upstream?

Origin : Its the personal copy of a repo that you have cloned or your own repo that you have created . You have the write access and you can push your changes

Upstream : its the source , the repo that you copied from .

For eg: 90DaysofDevops Challenge created by TWS is the upstream and origin will be our personal copy of repo that we forked from TWS 90DaysofDevops .

Task 4: Pull from GitHub



Pull that change to your local repo

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git switch master
Switched to branch 'master'
ubuntu@ip-172-31-30-155:~/devops-git-practice$ ls -ltr
ls: cannot access 's-ltr': No such file or directory
ubuntu@ip-172-31-30-155:~/devops-git-practice$ ls -ltr
total 12
-rwxrwxr-x 1 ubuntu ubuntu 33 Mar 12 15:16 hello-world.sh
-rwxrwxr-x 1 ubuntu ubuntu 67 Mar 12 15:21 greet.sh
-rw-rw-r-- 1 ubuntu ubuntu 52 Mar 12 15:23 index.html
ubuntu@ip-172-31-30-155:~/devops-git-practice$ cat hello-world.sh
#!/bin/bash

echo "Hello World"
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git pull origin master
From github.com:samsrajyt/devops-git-practice-remote
 * branch                master       -> FETCH_HEAD
Updating 3ce969d..539809b
Fast-forward
 hello-world.sh | 1 +
 1 file changed, 1 insertion(+)
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |
```

What is the difference between git fetch and git pull?

git pull (git fetch + git merge)

It pulls the remote changes and merges them into the local repo and the changes are visible in local.

git fetch

```
ubuntu@ip-172-31-30-155:~/devops-git-practice$ cat hello-world.sh
#!/bin/bash

echo "Hello World"
echo "Added one more line"
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git fetch origin master
From github.com:samsrajyt/devops-git-practice-remote
* branch          master      -> FETCH_HEAD
ubuntu@ip-172-31-30-155:~/devops-git-practice$ git diff HEAD FETCH_HEAD
diff --git a/hello-world.sh b/hello-world.sh
index b61e798..4c8f873 100755
--- a/hello-world.sh
+++ b/hello-world.sh
@@ -3,3 +3,4 @@
 
 echo "Hello World"
 echo "Added one more line"
+echo "Added one more line"
ubuntu@ip-172-31-30-155:~/devops-git-practice$
```

Breaking Down the Parts

- *** branch:** This indicates that a branch was successfully contacted on the remote server (like GitHub).
- **master:** This is the name of the branch on the remote server that Git just looked at.
- **-> FETCH_HEAD:** This is the most important part. **FETCH_HEAD** is a short-lived reference that keeps track of what was just fetched from the remote repository.

Task 5: Clone vs Fork


```

ubuntu@ip-172-31-30-155:~/devops-git-practice$ git clone https://github.com/raxx21/rajesh-portfolio.git
Cloning into 'rajesh-portfolio'...
remote: Enumerating objects: 180, done.
remote: Counting objects: 100% (61/61), done.
remote: Compressing objects: 100% (31/31), done.
remote: Total 180 (delta 32), reused 30 (delta 30), pack-reused 119 (from 1)
Receiving objects: 100% (180/180), 37.75 MiB | 41.25 MiB/s, done.
Resolving deltas: 100% (55/55), done.
ubuntu@ip-172-31-30-155:~/devops-git-practice$ ls -l
total 16
-rwxrwxr-x 1 ubuntu ubuntu  67 Mar 12 15:21 greet.sh
-rwxrwxr-x 1 ubuntu ubuntu  60 Mar 13 11:03 hello-world.sh
-rw-rw-r-- 1 ubuntu ubuntu  52 Mar 12 15:23 index.html
drwxrwxr-x 5 ubuntu ubuntu 4096 Mar 13 11:34 rajesh-portfolio
ubuntu@ip-172-31-30-155:~/devops-git-practice$ |

```

This screenshot shows the GitHub repository page for `raxx21/rajesh-portfolio`. The repository is public and has 164 forks and 285 stars. The main branch is selected, and the file list shows `public`, `src`, `.gitignore`, and `LICENSE`. The `About` section on the right indicates that there is no description, website, or topics provided.

This screenshot shows the 'Create a new fork' page for the `raxx21/rajesh-portfolio` repository. The page prompts the user to create a fork, explaining that a fork is a copy of the repository. The 'Owner' is set to `samsrajyt` and the 'Repository name' is `rajesh-portfolio`. The 'Description' field is empty. The 'Copy the main branch only' checkbox is checked. The 'Create fork' button is visible at the bottom right.

This screenshot shows the GitHub repository page for `samsrajyt/rajesh-portfolio`, which is a fork of `raxx21/rajesh-portfolio`. The page shows the repository is up to date with the upstream repository. The file list and the `About` section are visible, showing that the fork has 0 stars and 0 forks.

What is the difference between clone and fork?

clone is used to get a copy of remote repo onto your local machine and use this when you have **permission** to work on a project (like your own repo or a company project).

fork is used to copy a remote repo to another remote repo and Use this when you **don't have permission** to change the original code (like contributing to Open Source). You fork it to your own account first, then clone *your* fork to your machine

When would you clone vs fork?

clone - on local machine

Fork - on remote repo like Github

After forking, how do you keep your fork in sync with the original repo?

We can use the sync fork option to do that.